



Automated single-platform telomere-to-telomere *de novo* genome assembly for human, plant, and animal genomes

Advances in nanopore read accuracy, as well as read correction and assembly algorithms, enable routine T2T assembly for platinum-standard references and population-scale pangenome analysis

More information at: www.nanoporetech.com and nanoporetech.com/resource-centre

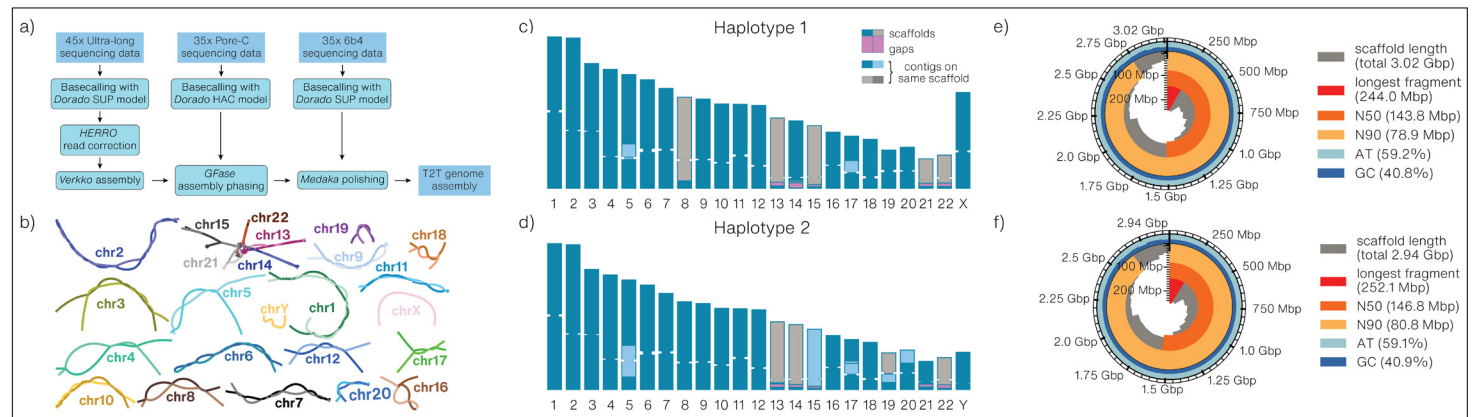


Fig. 1 a) Telomere-to-telomere assembly pipeline b) bandage plot of Verikko assembly graph c) and d) karyoplots of haplotypes 1 and 2 e) and f) assembly statistics for haplotypes 1 and 2

Analysis of ultra-long and Pore-C sequencing data with the *HERRO*, *Verikko*, and *GFase* algorithms yields the most complete automated human genome assembly yet produced

Telomere-to-telomere (T2T) assemblies provide the most information possible about an organism's genotype, allowing for analysis of previously inaccessible regions of the genome, like centromeric satellites and other large/high copy repeat regions. The previous state-of-the-art for generating T2T assemblies required both highly accurate reads like nanopore duplex reads as well as ultra-long simplex reads; however, advances in raw read accuracy and the new *HERRO* correction algorithm brings the corrected accuracy of simplex reads on par with duplex reads. Assembly of 45x of *HERRO*-corrected ultra-long reads (N50 = 100kbp) using the *Verikko* assembler yielded 20 large graph components corresponding to the 19 metacentric human chromosomes plus the acrocentric chromosomes (Fig. 1a, b). Phasing of this assembly graph with 35x of Pore-C data using the *GFase* algorithm followed by chromosome reconstruction within *Verikko* yielded 30 T2T contigs and five additional T2T scaffolds (Fig. 1a, c, d). Haplotype 1 plus the X chromosome contained 3.02 Gbp assembled sequence, with a scaffold N50 of 143.8 Mbp (Fig. 1e), while Haplotype 2 plus the Y chromosome contained 2.94 Gbp assembled sequence, with a scaffold N50 of 146.8 Mbp (Fig. 1f).

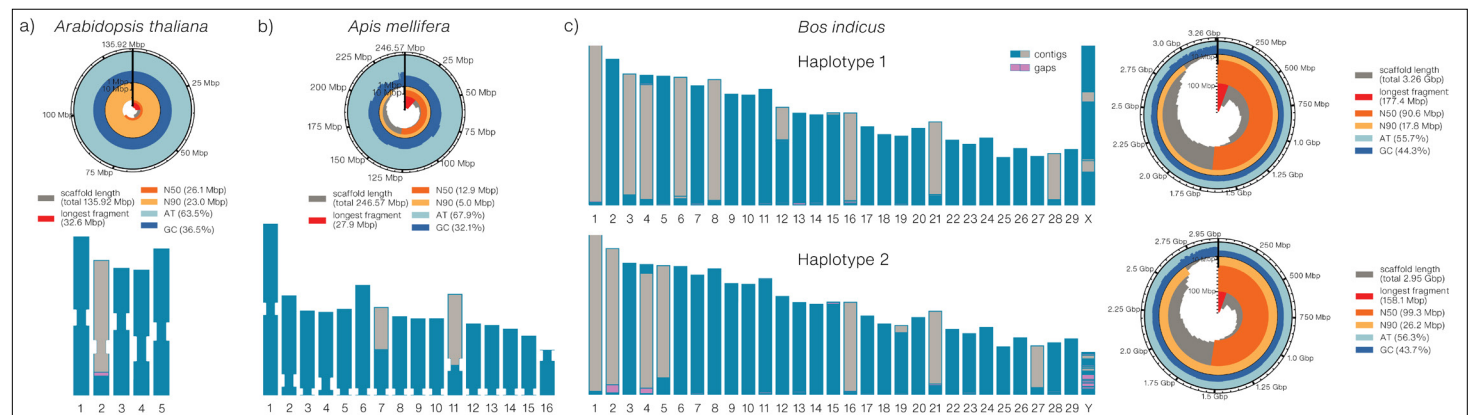


Fig. 2 a) Assembly statistics and karyplot of *Arabidopsis thaliana* b) assembly statistics and karyplot of *Apis mellifera* c) assembly statistics and karyoplots for each haplotype of *Bos indicus*

Accurate and complete assemblies for both small and large genomes from a diverse range of organisms with highly scalable T2T assembly workflows

Gapless telomere-to-telomere assemblies can benefit plant and animal genomic applications by providing nearly perfect reference assemblies, as well as facilitating the discovery of the most complete genomic variants. Data requirements and assembly workflows can be tailored to the genome size and ploidy of the organism being analysed. We assembled 20 Gbp of nanopore duplex data from *Arabidopsis thaliana* with the analysed assembler, generating an assembly with three T2T chromosomes and the remaining two broken only at rDNA arrays (Fig. 2a). Both the honeybee (*Apis mellifera*) and zebu cattle (*Bos indicus*) have primarily or exclusively acrocentric chromosomes, respectively, limiting assembly of most chromosomes. Nonetheless, we assembled six of the 16 honeybee chromosomes T2T using 20 Gbp analysed-corrected simplex reads with the analysed assembly pipeline (Fig. 2b). For zebu cattle, we assembled 140 Gbp of duplex data, 170 Gbp of ultra-long data, and 100 Gbp of Pore-C data using the analysed + analysed assembly pipeline, yielding a highly complete and contiguous assembly with five chromosomes assembled as T2T contigs (Fig. 2c). For both the honeybee and the zebu cattle, the majority of the remaining chromosomes were assembled telomere-to-centromere.